

NE 523: Homework 2

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1 Definition of Fluxes and Currents

1. For a beam of intensity I in $+\hat{x}$, we know the current in $+\hat{x}$ is equal to the beam intensity.

$$J_x^+ = I \quad (1)$$

2. As the beam is only traveling in the positive x direction in a collisionless void, so no neutrons can be traveling in the $-\hat{x}$ direction.

$$J_x^- = 0 \quad (2)$$

3. There is no current in the y or z directions, so the current can be written as:

$$\vec{J} = \langle J_x^+ + J_x^-, 0, 0 \rangle = \langle I, 0, 0 \rangle \quad (3)$$

4. Similarly, we know the scalar flux is defined as the net number of neutrons traveling through an area in a given time. Because there is no current in the negative x direction, the current in the positive x direction and the scalar flux are equal.

$$\phi = J_x^+ = I \quad (4)$$

5. We know the scalar flux is found with the following integral.

$$\phi = \int_{4\pi} d\Omega \psi \quad (5)$$

6. Which we rewrite as:

$$\phi = \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \psi = 2\pi \int_{-1}^1 d\mu \psi \quad (6)$$

7. Because the current is only has a component in the positive x direction, we know the angular flux can be written as some magnitude A with the correct units and a delta function when $\mu = 1$.

$$\phi = 2\pi \int_{-1}^1 d\mu A \delta(\mu - 1) = 2\pi A \int_{-1}^1 d\mu \delta(\mu - 1) = 2\pi A \quad (7)$$

8. We solve the integral:

$$\phi = 2\pi A(1) = 2\pi A \quad (8)$$

9. When the angular flux is defined, it's magnitude is equal to $A = I/2\pi$ with $\phi = I$. We can write the angular flux as keeping in mind the angular flux is zero except in the positive x direction.

$$\psi = \frac{I}{2\pi} \delta(1 - \mu) \quad (9)$$

2 Solving the Transport Equation in Slab Geometry

1. First, start with the transport equation for a slab with an isotropic source.

$$\hat{\Omega} \cdot \nabla \psi(\mathbf{r}, \hat{\Omega}, E) = \frac{s(E)}{4\pi} \quad (10)$$

2. Since the slab is infinite and homogeneous with changes in y and z, we know $\frac{\partial \psi}{\partial y} = \frac{\partial \psi}{\partial z} = 0$ and Ω becomes μ .

$$\mu \frac{d\psi(x, \hat{\Omega}, E)}{dx} + \Sigma_t \psi(x, \hat{\Omega}, E) = \frac{s(E)}{4\pi} \quad (11)$$

3. Rearranging to isolate each variable:

$$\frac{1}{\Sigma_t \psi(\mathbf{r}, \hat{\Omega}, E) - \frac{s(E)}{4\pi}} d\psi(\mathbf{r}, \hat{\Omega}, E) = \frac{-dx}{\mu} \quad (12)$$

4. Integrate both sides:

$$\int \frac{1}{\Sigma_t \psi(\mathbf{r}, \hat{\Omega}, E) - \frac{s(E)}{4\pi}} d\psi(\mathbf{r}, \hat{\Omega}, E) = \int \frac{-dx}{\mu} \quad (13a)$$

$$\frac{\ln \left(\Sigma_t \psi - \frac{s(E)}{4\pi} \right)}{\Sigma_t} = \frac{-x}{\mu} + C_1 \quad (13b)$$

5. Rearrange:

$$\ln \left(\Sigma_t \psi - \frac{s(E)}{4\pi} \right) = \frac{-x \Sigma_t}{\mu} + C_1 \Sigma_t \quad (14)$$

6. Take the exponential keeping in mind the C_1 term was folded into the new constant:

$$\Sigma_t \psi - \frac{s(E)}{4\pi} = C_2 \exp\left(\frac{-x\Sigma_t}{\mu}\right) \quad (15)$$

7. Isolate ψ , note the constant changes when we fold in Σ_t :

$$\psi = C_3 \exp\left(\frac{-x\Sigma_t}{\mu}\right) + \frac{s(E)}{4\pi} \quad (16)$$

8. When $\mu > 0$, we know $C_3 = 0$ because flux is finite. We also know, for $\mu < 0$, there is no incoming flux at $x = 0$. Because the incoming flux is 0, we know the exponential term and source term cancel at $x = 0$, so the constant $C_3 = \frac{s(E)}{4\pi\Sigma_t}$.

9. This gives rise to the following form of ψ

$$\psi(x, \mu, E) = \begin{cases} \frac{s(E)}{4\pi\Sigma_t}, & \mu \geq 0 \\ \frac{s(E)}{4\pi\Sigma_t} \left(1 - \exp\left(\frac{-x\Sigma_t}{\mu}\right)\right), & \mu < 0 \end{cases} \quad (17)$$

10. Next, we evaluate the angular flux at $x = 0$:

$$\psi(x, \mu, E) = \begin{cases} \frac{s(E)}{4\pi\Sigma_t}, & \mu \geq 0 \\ 0, & \mu < 0 \end{cases} \quad (18)$$

11. Finally, the current vector at $x = 0$:

$$\mathbf{J}(0, E) = \int_{4\pi} d\hat{\Omega} \psi = \left(\frac{s(E)}{2\Sigma_t}, 0, 0\right) \quad (19)$$

3 Inner Products and Orthogonality

We know two functions are orthogonal if $\int d\mu P_i P_j$ is 0 when $i \neq j$ and non-zero when $i = j$. Here, μ is the measure over which the functions are orthogonal. Functions are orthonormal if they are orthogonal and the inner product of the function with itself equals one.

3.1 Zeroth Polynomial

1. The inner product of the zeroth polynomial and the zeroth polynomial is one.

$$\int_{-1}^1 dx P_0 P_0 = \int_{-1}^1 dx \frac{1}{2} = \frac{1}{2} [x]_{-1}^1 = 1 \quad (20)$$

2. The inner product of the zeroth polynomial and the second polynomial is zero.

$$\int_{-1}^1 dx P_0 P_1 = \int_{-1}^1 dx \frac{\sqrt{3}}{2} x = \frac{\sqrt{3}}{2} \left[\frac{1}{2} x^2 \right]_{-1}^1 = 0 \quad (21)$$

3. The inner product of the zeroth polynomial and the second polynomial is zero.

$$\int_{-1}^1 dx P_0 P_2 = \int_{-1}^1 dx \frac{\sqrt{5}}{2} \left(\frac{3x^2 - 1}{2} \right) = \frac{\sqrt{5}}{8} \{ x^3|_{-1}^1 - x|_{-1}^1 \} = 0 \quad (22)$$

3.2 First Polynomial

1. The inner product of the first polynomial and the zeroth polynomial is equal to the inner product of the zeroth polynomial and the first polynomial.

$$\int_{-1}^1 dx P_1 P_0 = \int_{-1}^1 dx P_0 P_1 = 0 \quad (23)$$

2. The inner product of the first polynomial and the first polynomial is one.

$$\int_{-1}^1 dx P_1 P_1 = \int_{-1}^1 dx \frac{3}{2} x^2 = \frac{3}{2} \left(\frac{1}{3} x^3|_{-1}^1 \right) = 1 \quad (24)$$

3. The inner product of the first polynomial and the second polynomial is zero.

$$\int_{-1}^1 dx P_1 P_2 = \int_{-1}^1 dx \frac{\sqrt{15}}{4} (3x^3 - x) = \frac{\sqrt{15}}{4} \left\{ \left[\frac{3}{4} x^4 \right]_{-1}^1 - \left[\frac{1}{2} x^2 \right]_{-1}^1 \right\} = 0 \quad (25)$$

3.3 Second Polynomial

1. The inner product of the second polynomial and the zeroth polynomial is equal to the inner product of the zeroth polynomial and the second polynomial.

$$\int_{-1}^1 dx P_2 P_0 = \int_{-1}^1 dx P_0 P_2 = 0 \quad (26)$$

2. The inner product of the second polynomial and the first polynomial is equal to the inner product of the second polynomial and the second polynomial.

$$\int_{-1}^1 dx P_2 P_1 = \int_{-1}^1 dx P_1 P_2 = 0 \quad (27)$$

3. The inner product of the second polynomial and the second polynomial is one.

$$\int_{-1}^1 dx P_2 P_2 = \int_{-1}^1 dx \frac{5}{8} (3x^2 - 1)^2 = \frac{5}{8} \left\{ \left[\frac{9}{5} x^5 \right]_{-1}^1 - [2x^3]_{-1}^1 + [x]_{-1}^1 \right\} = \frac{5}{8} \left[\frac{18}{5} - 4 + 2 \right] = 1 \quad (28)$$

The first three Legendre polynomials are orthonormal.

4 Projection on Orthogonal Basis and Approximation

1. Using the polynomials (P_i) from problem 3, we can write an approximation where $f_i = \int_{-1}^1 dx f(x) P_i(x)$:

$$\tilde{f}(x) = \sum_{i=0}^2 f_i P_i(x) \quad (29)$$

2. We will calculate each coefficient in the approximation, then plot the approximations next to the exact function.

4.1 $3x + 7$

$$f_0 = \int_{-1}^1 dx \frac{1}{\sqrt{2}} (3x + 7) = 7\sqrt{2} \quad (30a)$$

$$f_1 = \int_{-1}^1 dx \sqrt{\frac{3}{2}} x (3x + 7) = \sqrt{6} \quad (30b)$$

$$f_2 = \int_{-1}^1 dx \sqrt{\frac{5}{2}} \left(\frac{3x^2 - 1}{2} \right) (3x + 7) = 0 \quad (30c)$$

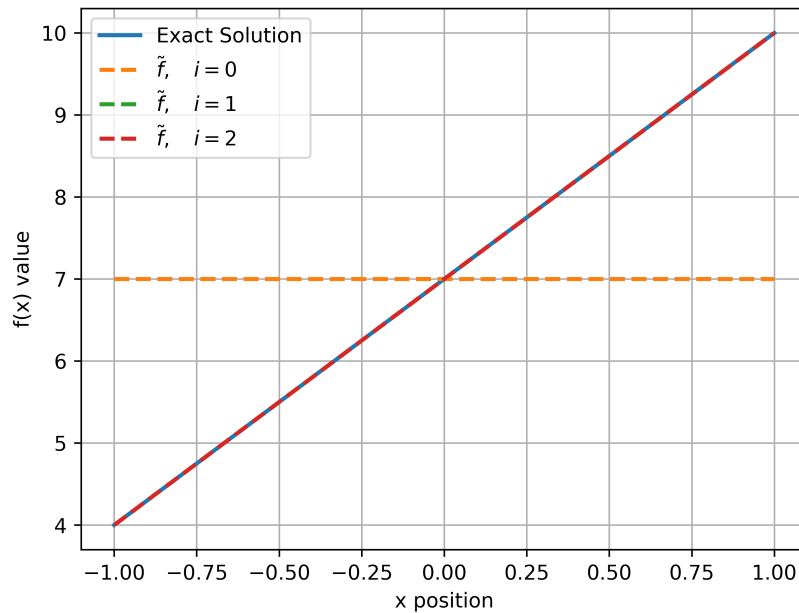


Figure 1: Approximating f_a using Legendre Polynomials

4.2 $\exp x$

$$f_0 = \int_{-1}^1 dx \frac{1}{\sqrt{2}} \exp x = \frac{1}{\sqrt{2}} [\exp 1 - \exp(-1)] \quad (31a)$$

$$f_1 = \int_{-1}^1 dx \sqrt{\frac{3}{2}} x \exp x = \sqrt{6} \exp(-1) \quad (31b)$$

$$f_2 = \int_{-1}^1 dx \sqrt{\frac{5}{2}} \left(\frac{3x^2 - 1}{2} \right) \exp x = \sqrt{\frac{5}{2}} [\exp(1) - 7 \exp(-1)] \quad (31c)$$

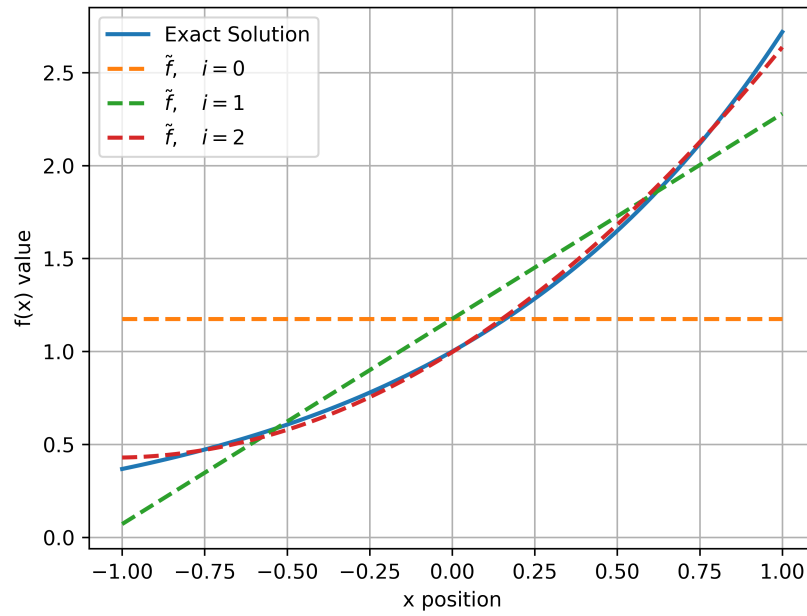


Figure 2: Approximating f_b using Legendre Polynomials

4.3 $\exp 5x$

$$f_0 = \int_{-1}^1 dx \frac{1}{\sqrt{2}} \exp 5x = \frac{1}{5\sqrt{2}} [\exp 5 - \exp(-5)] \quad (32a)$$

$$f_1 = \int_{-1}^1 dx \sqrt{\frac{3}{2}} x \exp 5x = \frac{\sqrt{6}}{25} [2 \exp(5) - 3 \exp(-5)] \quad (32b)$$

$$f_2 = \int_{-1}^1 dx \sqrt{\frac{5}{2}} \left(\frac{3x^2 - 1}{2} \right) \exp 5x = \frac{1}{5^{5/2}\sqrt{2}} [13 \exp(5) - 43 \exp(-5)] \quad (32c)$$

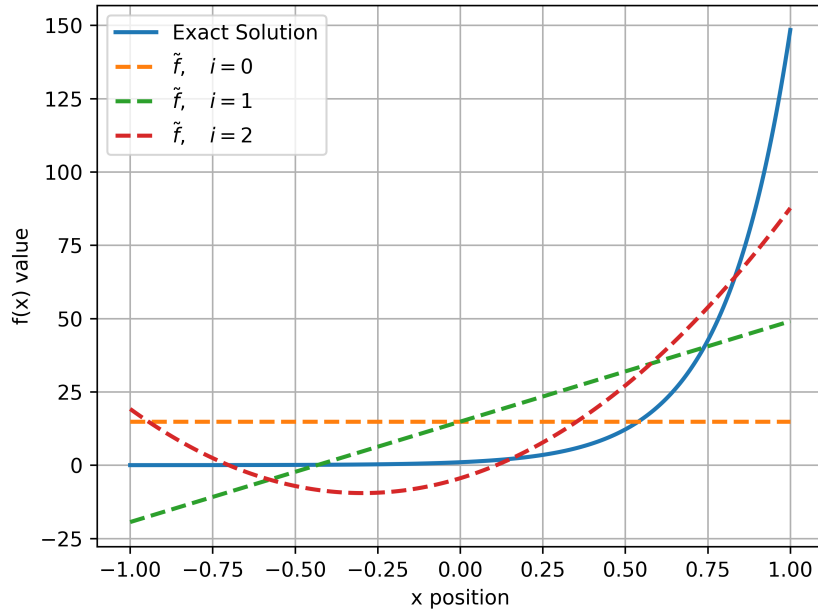


Figure 3: Approximating f_c using Legendre Polynomials